

Introduction to smooth geometry

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Abstract

These are notes for a talk given to the [Boston University Mathematics Association](#) in the Fall 2026 semester. We take a somewhat less rigorous approach to the subject, and focus on breadth and intuition. We introduce manifolds, tangent spaces and the tangent bundle, vector fields the exterior derivative, differential forms, and integration.

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1 Manifolds

motivate manifolds in the obvious way

2 The tangent space

So far, we have been very interested in functions $\mathbb{R}^n \rightarrow \mathbb{R}$. Thus, it makes sense to look at functions $M \rightarrow \mathbb{R}$, for M some manifold. To start doing calculus on manifolds, we should start by discussing the notion of a derivative for such a function, since derivatives are historically easiest to study. In calculus, given a function $f : \mathbb{R} \rightarrow \mathbb{R}$, we start by defining the derivative at a *point* $x \in \mathbb{R}$ using the limit

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

In multivariate calculus, we started by defining the partial derivative $\partial_i f$ of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at a point $x \in \mathbb{R}^n$ using a similar limit definition. Thus, we should start by attempting to define the derivative of a function $f : M \rightarrow \mathbb{R}$ at a point.

Let M be an n -manifold, let $f : M \rightarrow \mathbb{R}$ be any function, and let $p \in M$ be a point. We can use the coordinates of M to obtain a notion of a derivative. Choose a chart (U, x) such that $p \in U$. Recall that this means x is an isomorphism $x : U \cong V \subseteq \mathbb{R}^n$ to some open subset V of $\mathbb{R}^n$. Recall that this allows us to write any point $q \in U$ using coordinates $x(q) = (x^1(q), \dots, x^n(q))$, which we usually just write as $(x^1, \dots, x^n)$ for simplicity. This means that if we only look at the values of f on U , then we can write f as a function in coordinates

$$f(x) : \mathbb{R}^n \rightarrow \mathbb{R}.$$

This should be good enough to define a derivative at p , since the derivative only depends on the value of f in an infinitesimally small region near p .

Definition 2.1. With notation as above, define the *partial derivative of f with respect to x^i at p* to be the value

$$\left. \frac{\partial f(x)}{\partial x^i} \right|_p,$$

where $f(x)$ represents the function f in coordinates, and $|_p$ denotes “evaluation at $x(p)$ ”.

Notation 2.2. We often write $\partial_i|f$ or $\frac{\partial}{\partial x^i}|_p f$ to denote the partial derivative of f evaluated at p .

If we want to take multiple partial derivatives of f at p , then we just take the partial derivatives before evaluating at p . For example,

$$\left. \frac{\partial}{\partial x^i} \right|_p \left. \frac{\partial}{\partial x^j} \right|_p f = \left. \frac{\partial^2 f}{\partial x^i \partial x^j} \right|_p.$$

For this reason, we usually only write one evaluation bar. We will usually use notation like

$$\left(\frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j} \right) \Big|_p f \text{ or } \frac{\partial^2}{\partial x^i \partial x^j} \Big|_p f.$$

Definition 2.3. Let M be a manifold, and let $U \subseteq M$ be an open subset. Let $f : U \rightarrow \mathbb{R}$ be any function. We say f is *smooth at $p \in M$* if all possible combinations of partial derivatives of f at p exist. The set of all smooth functions $U \rightarrow \mathbb{R}$ is denoted $C_M^\infty(U)$.

If M and N are manifolds, then using the coordinates on both M and N simultaneously, we can provide the same definition for smooth functions $f : M \rightarrow N$.

Although we could work with functions that are not smooth (eg. continuously differentiable functions), smooth functions are much more interesting, are much better behaved, are much more critical to developing the theory, and are still abundant in nature. Thus, we will focus extensively on smooth functions. We will highlight two of these properties right now. If $U \subseteq M$ and $f, g : U \rightarrow \mathbb{R}$ are smooth, then define the functions $f + g, fg : U \rightarrow \mathbb{R}$ by

$$(f + g)(p) = f(p) + g(p), \quad (fg)(p) = f(p)g(p).$$

Then both $f + g$ and fg are smooth functions. In other words, $C_M^\infty(U)$ is a *ring*. For those with knowledge of sheaves, C_M^∞ is a sheaf of rings.

There’s a particular property about partial derivatives at a point p which will help us significantly moving forward: partial derivatives at p form a *vector space*.

Proposition 2.4. *Let M be an n -manifold, and let $p \in M$. Let (U, x) be a chart containing p . Then the partial derivatives $\{\partial_i|_p\}_{1 \leq i \leq n}$ form a basis for a vector space.*

Proof. In order for partial derivatives to form a vector space, we need to know how to add partial derivatives, and how to multiply them by a scalar $c \in \mathbb{R}$. A partial derivative is just a function which takes in smooth functions $C_M^\infty(U)$ (or defined on any subset containing p), and sends them to $\mathbb{R}$. Thus to define $\partial_i|_p + \partial_j|_p$, we just need to know how to evaluate it on a function. Define

$$(\partial_i|_p + \partial_j|_p)f = \partial_i|_p f + \partial_j|_p f.$$

This is a pretty simple definition. The definition for multiplying by a scalar is similarly simple:

$$(c\partial_i|_p)f = c(\partial_i|_p f).$$

The vector space axioms are fairly routine to check, and we omit them for conciseness. The proof that the ∂_i ’s are linearly independent is left as an exercise (hint: construct a function $f^i : \mathbb{R}^n \rightarrow \mathbb{R}$ and $p \in \mathbb{R}^n$ such that $\partial_j|_p f^i = 1$ if and only if $i = j$, and 0 otherwise. Using the coordinates $x : U \cong V \subseteq \mathbb{R}^n$, argue that this is sufficient). ■

The vector space constructed in proposition 2.4 is invariant under choice of coordinates. Suppose (V, y) is another set of coordinates, and let φ be the transition function. Recall that this means $y = \varphi(x)$. Let $f : V \rightarrow \mathbb{R}$ be a smooth function, and suppose that I can write f in the coordinates y . Then since $y = \varphi(x)$,

we can write f in the coordinates x as well using $f(y) = f(\varphi(x))$. We can now use the chain rule to calculate the partial derivatives of f in the x^i direction:

$$\left. \frac{\partial f}{\partial x^i} \right|_p = \left. \frac{\partial f(\varphi(x))}{\partial x^i} \right|_p = \sum_j \left. \frac{\partial f}{\partial y^j} \right|_p \left. \frac{\partial \varphi^j}{\partial x^i} \right|_p = \left(\sum_j \left. \frac{\partial \varphi^j}{\partial x^i} \right|_p \left. \frac{\partial}{\partial y^j} \right|_p \right) f.$$

Note that the component

$$\left. \frac{\partial \varphi^j}{\partial x^i} \right|_p$$

is just a number, i.e. a *scalar*. Thus,

$$\sum_j \left. \frac{\partial \varphi^j}{\partial x^i} \right|_p \left. \frac{\partial}{\partial y^j} \right|_p$$

is an element of the vector space given by partial derivatives in the coordinates y^i . Thus, we see that the transition function φ gives us a way of transitioning from the vector space of partial derivatives with respect to x^i , to the vector space of partials with respect to y^i . This can also be done in the reverse direction, meaning that these vector spaces are *isomorphic*. In other words, this vector space doesn't depend on the choice of coordinates, since we can always use the transition function to switch between coordinates.

Definition 2.5. The vector space constructed in proposition 2.4 is denoted $T_p M$, and is called the *tangent space to p* .

Now, we can define a total derivative of a function at a point p . Recall that in single variable calculus, the total derivative of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ at a point p is just the number $f'(p)$. In multivariable calculus, our best notion of a total derivative was the *gradient vector*, meaning that the total derivative at a point p of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a *vector* $\nabla f(p) \in \mathbb{R}^n$. If we have a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we can upgrade the notion of a total derivative to the *Jacobian matrix*: writing $f(x) = (f^1(x), \dots, f^m(x))$, define the Jacobian matrix at p , denoted $Df(p)$, as the matrix of partial derivatives

$$Df(p) = \begin{pmatrix} \partial_1|_p f^1 & \dots & \partial_n|_p f^1 \\ \vdots & \ddots & \vdots \\ \partial_1|_p f^m & \dots & \partial_n|_p f^m \end{pmatrix}.$$

Essentially, the derivative of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is an $m \times n$ matrix.

Using this intuition, we can define the derivative of a smooth function $f : M \rightarrow N$ between *manifolds*. Let M be m -dimensional, and let N be n -dimensional. Let $p \in M$, let (U, x) be a chart for p on M , and let (V, y) be a chart for $f(p)$ on N . Then using these coordinates, we can write f as a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$. We can then define the *differential of f at p* , denoted df_p , to be the Jacobian matrix

$$df_p = Df(p).$$

This definition has a problem: we want our definitions to be coordinate-independent. That is, if I have different coordinates, I should get the same answer. But that isn't the case here; changing coordinates changes the matrix itself. However, we can fix this by viewing df_p as a *linear transformation*. Recall that a linear transformation can be given by different matrices when we change the basis, and recall that coordinates give us a basis for $T_p M$ and $T_{f(p)} N$. Thus, we will view df_p as a linear transformation $T_p M \rightarrow T_{f(p)} N$, whose *basis representation* is $Df(p)$.

Let's unpack what this means. Let $V \in T_p M$, and use the basis to write

$$V = \sum_i c_i \partial_i|_p, \quad c_i \in \mathbb{R} \forall i.$$

Let's treat the $\partial_i|_p$ as basis vectors, and rewrite this as a column vector:

$$V = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

Then we can calculate

$$(df_p)V = \begin{pmatrix} \partial_1|_p f^1 & \cdots & \partial_n|_p f^1 \\ \vdots & \ddots & \vdots \\ \partial_1|_p f^m & \cdots & \partial_n|_p f^m \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} c_1 \partial_1|_p f^1 + \cdots + c_n \partial_n|_p f^1 \\ \vdots \\ c_1 \partial_1|_p f^m + \cdots + c_n \partial_n|_p f^m \end{pmatrix}.$$

If we rewrite this column vector using the basis ∂_j (here we take the partials wrt y), we have

$$= \sum_i \sum_j c_j \left. \frac{\partial f^i}{\partial x^j} \right|_p \left. \frac{\partial}{\partial y^i} \right|_p.$$

Using this formula, we can calculate that

$$(df_p)(\partial_j|_p) = \sum_i \left. \frac{\partial f^i}{\partial x^j} \right|_p \left. \frac{\partial}{\partial y^i} \right|_p.$$

Let's look at the special case of a function $f : M \rightarrow \mathbb{R}$. Then we note that $T_p\mathbb{R}$ always has a basis given by $\partial_x|_p$, since $\mathbb{R}$ has a global set of coordinates given by normal coordinates (we call this a *global chart*). This means that we can write

$$(df_p)(\partial_i|_p) = \left. \frac{\partial f}{\partial x^i} \right|_p \left. \frac{\partial}{\partial x} \right|_p.$$

We will come back to this calculation. For now, we will close with the following remark about tangent spaces.

If f is a function $f : \mathbb{R} \rightarrow \mathbb{R}$, then one definition of $f'(x)$ is the *slope of the line tangent to f at the point x* . If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, then one definition of ∇f is the *tangent vector to the graph of f with maximal slope*. Continuing this trend, there is a way of thinking about df as a vector tangent to f . In particular, notice that $\partial_i|_p$ produces tangent vectors, and thus we can think of $\partial_i|_p$ as a tangent vector *itself*. The vector $\partial_i|_p$ is the vector tangent to p which points in the x^i -direction. This geometric interpretation of the algebraic object $\partial_i|_p$ turns out to be well-motivated; there is a theorem proving that this interpretation works. In this way, we think of the entire tangent space $T_p M$ as the space of vectors that are tangent to p . This interpretation will be useful.

3 The tangent bundle

I don't know about you guys, but I'm tired of carrying around the "evaluation at $p \in M$ " symbol. If we have coordinates on an open set U , we can do derivatives anywhere on U using those coordinates, so why do we have to carry around a symbol for derivatives at a single point? Additionally, we have been working extensively with coordinates. But coordinates can be messy, and are not necessarily defined everywhere. It would be nice to have some coordinate-independent properties of derivatives. The *tangent bundle* TM will solve all of these problems.

The idea of the tangent bundle is exactly what the name says: we bundle together all of the tangent spaces into a new manifold. More precisely, the definition is as follows.

Definition 3.1. Let M be an n -manifold. Define the *tangent bundle* TM to be the space

$$TM = \{(p, v) \mid p \in M, v \in T_p M\}.$$

The explicit description of the smooth structure is not important, but we make two comments. First, TM is a $2n$ -manifold. Second, there is a function $\pi : TM \rightarrow M$ given by $\pi(p, v) = p$. This map is smooth.

4 Differential forms and integration

Consider dx^i for x^i some coordinate. Show that $dx^i(\partial_j) = \delta_{ij}$ under the identification $T_p\mathbb{R} \cong \mathbb{R}$. Introduce T^*M as the dual bundle, and define the pullback map. Introduce df as a section $\Gamma(-, T^*M) =: \Omega^1$. Write $d : C^\infty(U) \rightarrow \Omega^1(U)$. Discuss $\mathbb{R}$ -linearity and $d(fg) = (df)g + f dg$. Write a general 1-form as

$(f dx^i)_p = f(p) dx^i|_p$ in coordinates. Argue that dx^i is indeed a differential in the direction x^i : a derivative is the operator giving us infinitesimal changes, and so this is like the length of that operator, which is infinitesimal... idk. Consider a line integral

$$\int_{\gamma} f dx^i = \int_0^1 \gamma^*(f dx^i) = \int_0^1 f(\gamma(t)) \gamma'(t) dt.$$

Finally, argue using orientations that we should work in $\Gamma(-\Lambda^\bullet T^*M)$. Define

$$d\omega = \sum_i \frac{\partial \omega_I}{\partial x^i} dx^i \wedge dx^I.$$

Argue $d^2 = 0$. Note that this yields the de Rham complex, and note the fascinating quasi-isomorphism to the singular cochain complex, and homotopy-invariance of $\Omega^\bullet$. Given a top form ω , define

$$\int_U \omega.$$

Note that some manifolds (Riemannian manifolds) have a canonical top form, called vol. We then define the integral of $f : M \rightarrow \mathbb{R}$ to be

$$\int_U f \text{vol}.$$

If M doesn't have a canonical top form, then it makes less sense to think about forms as measures, and more sense to think about them as the different things you integrate (since the interesting thing probably is the diff measures u get). Write Stokes' theorem

$$\int_M d\omega = \int_{\partial M} \omega.$$

Use this to prove the fundamental theorem of calculus. Note that this subsumes the fundamental theorem of line integrals, Stokes' theorem for vector fields, and Gauss' theorem. If time, remark on sheaves to motivate AG.